



United International University

Department of Computer Science and Engineering

BSCSE Curriculum

Updated on March 2026

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BSCSE Curriculum

Bachelor of Science in Computer Science and Engineering (CSE) primarily involves the study of a number of core courses which every CSE graduate should know and a significant number of courses from specialized areas. Core courses build the foundation and specialized courses prepare the students for the specific areas of Computer Science and Engineering. To understand the underpinning theory of the courses of Computer Science and Engineering, a number of courses on Mathematics and Basic Science have been felt mandatory to include in the syllabus. In addition, some social science, management, accounting, economics and communication-skills development related courses have been incorporated to make the syllabus a balanced and reasonably complete one. The objective of the undergraduate program in Computer Science and Engineering is to develop skilled and competent graduates to meet the current and future needs at home and abroad.

This curriculum is effective for the students of Summer 2026 and onwards.

The students from the previous batches can take the new courses and also do their majors from the new majors fulfilling the respective major's criteria.

Admission Requirements

Every applicant, without any exception, must fulfill the admission requirements as laid down by the university. Admission test and interview for admission into a trimester will be held as decided by the university.

A higher secondary certificate or its equivalent in science with mathematics and physics or other fields of study is the basic educational requirement.

Degree Requirements

The B.Sc. in CSE degree requirements will be as follows:

- (a) Completion of 141.0 credit hours
- (b) Completion of the final year design project with at least a 'C' grade
- (c) Passing of all courses individually and maintaining a minimum CGPA of 2.0

A specialization will be declared in one of the following areas if at least four courses are completed from the elective courses: AI & Data Science, Software Engineering, Network & Cybersecurity, Embedded System & Robotics, and Business System Engineering.

List of Courses

(A) Language (6 credits)

1	ENG 1011	English I	3.0
2	ENG 1013	English II	3.0

(B) General Education (17 credits)

Compulsory (8 credits)

1	SOC 2101	Society, Environment and Engineering Ethics	3.0
2	PMG 4101	Project Management	3.0
3	BDS 1201	History of the Emergence of Bangladesh	2.0

Optional (Any three: 9 credits)

1	ECO 4101	Economics	3.0
2	SOC 4101	Introduction to Sociology	3.0
3	ACT 2111	Financial and Managerial Accounting	3.0
4	IPE 3401	Industrial and Operational Management	3.0
5	GED 4003	Technology Entrepreneurship	3.0
6	PSY 2101	Psychology	3.0
7	BDS 2201	Bangladesh Studies	3.0
8	BAN 2501	Bangla	3.0
9	SOC 4301	Introduction to Anthropology	3.0
10	FLN 1101	Introduction to a Foreign Language	3.0
11	SWL 4101	Software and Law	3.0
12	BHV 2101	Introduction to Behavioral Science	3.0
13	URC 1101	Life Skill for Success	3.0

(C) Basic Sciences (7 credits)

1	PHY 2105	Physics	3.0
2	PHY 2106	Physics Laboratory	1.0
3	BIO 3105	Biology for Engineers	3.0

(D) Mathematics (12 credits)

1	MATH 1151	Fundamental Calculus	3.0
2	MATH 2183	Calculus and Linear Algebra	3.0
3	MATH 2201	Coordinate Geometry and Vector Analysis	3.0
4	MATH 2205	Probability and Statistics	3.0

(E) Other Engineering (10 credits)

List of Courses

1	EEE 2113	Electrical Circuits	3.0
2	EEE 2123	Electronics	3.0
3	EEE 2124	Electronics Laboratory	1.0
4	EEE 4261	Green Computing	3.0

(F) Core/Compulsory Courses (68 credits)

Programming (13 credits)

1	CSE 1110	Introduction to Computer Systems	1.0
2	CSE 1111	Structured Programming Language	3.0
3	CSE 1112	Structured Programming Language Laboratory	1.0
4	CSE 1115	Object Oriented Programming	3.0
5	CSE 1116	Object Oriented Programming Laboratory	1.0
6	CSE 2118	Advanced Object Oriented Programming Laboratory	1.0
7	CSE 4165	Web Programming	3.0

Hardware (11 credits)

1	CSE 1325	Digital Logic Design	3.0
2	CSE 1326	Digital Logic Design Laboratory	1.0
3	CSE 3313	Computer Architecture	3.0
4	CSE 4325	Microprocessors and Microcontrollers	3.0
5	CSE 4326	Microprocessors and Microcontrollers Laboratory	1.0

Logics and Algorithms (14 credits)

1	CSE 2213	Discrete Mathematics	3.0
2	CSE 2215	Data Structure and Algorithms I	3.0
3	CSE 2216	Data Structure and Algorithms I Laboratory	1.0
4	CSE 2217	Data Structure and Algorithms II	3.0
5	CSE 2218	Data Structure and Algorithms II Laboratory	1.0
6	CSE 2233	Theory of Computation	3.0

Software (8 credits)

1	CSE 3411	System Analysis and Design	3.0
2	CSE 3412	System Analysis and Design Laboratory	1.0
3	CSE 3421	Software Engineering	3.0
4	CSE 3422	Software Engineering Laboratory	1.0

Systems (22 credits)

List of Courses

1	CSE 4531	Computer Security	3.0
2	CSE 3521	Database Management Systems	3.0
3	CSE 3522	Database Management Systems Laboratory	1.0
4	CSE 4509	Operating Systems	3.0
5	CSE 4510	Operating Systems Laboratory	1.0
6	CSE 3711	Computer Networks	3.0
7	CSE 3712	Computer Networks Laboratory	1.0
8	CSE 3811	Artificial Intelligence	3.0
9	CSE 3812	Artificial Intelligence Laboratory	1.0
10	CSE 4889	Machine Learning	3.0

(G) Elective Courses (Any five: 15 credits)

i. Network and Cyber Security

1	CSE 4121	Security Monitoring	3.0
2	CSE 4759	Wireless and Cellular Communication	3.0
3	CSE 4123	Software and Application Security	3.0
4	CSE 4125	Ethical Hacking and Network Defense	3.0
5	CSE 4127	Cloud Security	3.0
6	CSE 4129	Digital Forensics	3.0
7	CSE 4131	Internet of Things (IoT) Security	3.0
8	CSE 3715	Data Communication	3.0
9	CSE 4199	Special Topics	3.0
10	CSE 4793	Advanced Network Services and Management	3.0
11	CSE 4783	Cryptography	3.0
12	CSE 4777	Network Security	3.0
13	CSE 4943	Web Application Security	3.0
14	CSE 4949	IT Audit: Concepts and Practice	3.0
15	CSE 3542	Industry Internship	3.0

ii. Software Engineering

1	CSE 4435	Software Architecture	3.0
2	CSE 4181	Mobile Application Development	3.0
3	CSE 4495	Software Testing and Quality Assurance	3.0
4	CSE 4587	Cloud Computing	3.0
5	CSE 4567	Advanced Database Management Systems	3.0
6	CSE 4413	Virtual Reality/Augmented Reality System Design	3.0
7	CSE 4611	Compiler Design	3.0
8	CSE 4123	Software and Application Security	3.0
9	CSE 4945	UI: Concepts and Design	3.0
10	CSE 4871	Data Analytics	3.0
11	CSE 4299	Special Topics	3.0
12	CSE 4133	Business Intelligence	3.0

List of Courses

13	CSE 4451	Human Computer Interaction	3.0
14	CSE 4485	Game Design and Development	3.0
15	CSE 4519	Distributed Systems	3.0
16	CSE 4523	Simulation and Modeling	3.0
17	CSE 4621	Computer Graphics	3.0
18	CSE 4601	Mathematical Analysis for Computer Science	3.0
19	CSE 4633	Basic Graph Theory	3.0
20	CSE 4655	Algorithm Engineering	3.0
21	CSE 4547	Multimedia Systems Design	3.0
22	CSE 4613	Computational Geometry	3.0
23	CSE 4949	IT Audit: Concepts and Practice	3.0
24	CSE 3542	Industry Internship	3.0
iii. Embedded System and Robotics			
1	CSE 4379	Real-time Embedded Systems	3.0
2	CSE 4337	Robotics	3.0
3	CSE 4399	Special Topics on Embedded System and Robotics	3.0
4	CSE 4327	VLSI Design	3.0
5	CSE 4853	Internet of Things (IoT)	3.0
6	CSE 4819	AI for Autonomous Systems	3.0
7	CSE 4813	Deep Learning	3.0
8	CSE 4815	Large Language Model and Generative AI	3.0
9	CSE 4329	Digital System Design	3.0
10	CSE 4397	Interfacing	3.0
11	CSE 3542	Industry Internship	3.0
iv. Business System Engineering			
1	CSE 4133	Business Intelligence	3.0
2	CSE 4891	Data Mining	3.0
3	CSE 4871	Data Analytics	3.0
4	CSE 4823	Data Driven Decision Making	3.0
5	CSE 4817	Big Data Analytics	3.0
6	CSE 4821	Digital Marketing	3.0
7	CSE 4945	UI: Concepts and Design	3.0
8	CSE 4451	Human Computer Interaction	3.0
9	CSE 4941	Enterprise Systems: Concepts and Practice	3.0
10	CSE 4765	Electronic Commerce	3.0
11	CSE 4949	IT Audit: Concepts and Practice	3.0
12	CSE 4499	Special Topics	3.0
13	CSE 3542	Industry Internship	3.0

v. AI and Data Science

1	CSE 4811	Natural Language Processing	3.0
2	CSE 4813	Deep Learning	3.0
3	CSE 4815	Large Language Model and Generative AI	3.0
4	CSE 4893	Introduction to Bioinformatics	3.0
5	CSE 4883	Digital Image Processing	3.0
6	CSE 4819	AI for Autonomous Systems	3.0
7	CSE 4891	Data Mining	3.0
8	CSE 4871	Data Analytics	3.0
9	CSE 4817	Big Data Analytics	3.0
10	CSE 4825	Speech Processing and Recognition	3.0
11	CSE 4949	IT Audit: Concepts and Practice	3.0
12	CSE 4599	Special Topics	3.0
13	CSE 3542	Industry Internship	3.0

vi. Final Year Design Project (6 credits)

1	CSE 4000A	Final Year Design Project - I	2.0
2	CSE 4000B	Final Year Design Project - II	2.0
3	CSE 4000C	Final Year Design Project - III	2.0

Summary of Courses

#	Group	Theory	Laboratory	Final Year De- sign Project	Total
1	Language	6.0	—	—	6.0
2	General Education	17.0	—	—	17.0
3	Basic Sciences	6.0	1.0	—	7.0
4	Mathematics	12.0	—	—	12.0
5	Other Engineering	9.0	1.0	—	10.0
6	Core/Compulsory Courses	54.0	14.0	—	68.0
7	Elective Courses	15.0	—	—	15.0
8	Final Year Design Project	—	—	6.0	6.0
Total		119.0	16.0	6.0	141.0

Course Equivalence

Main Course			Equivalent Course		
Course Code	Course Title	Cr.	Course Code	Course Title	Cr.
CSE 4853	Internet of Things	3.0	EEE 4523	Internet of Things - I	3.0
CSE 4379	Real-time Embedded Systems	3.0	EEE 4513	Real Time Embedded System Design	3.0
CSE 4337	Robotics	3.0	EEE 4515	Industrial Automation and Robotics	3.0
CSE 4327	VLSI Design	3.0	EEE 4121	VLSI Design	3.0
CSE 4399	Special Topics on Embedded System and Robotics	3.0	EEE 4519	Special Topics on Embedded System and Robotics	3.0
CSE 4821	Digital Marketing	3.0	MKT 4313	Digital Marketing	3.0
CSE 4765	Electronic Commerce	3.0	CSE 4763	Electronic Business	3.0
GED 4003	Technology Entrepreneurship	3.0	TEC 2499	Technology Entrepreneurship	3.0

Course Sequence

Trimester 1

Sl. No.	Course Code	Course Title	Credit Hr.	Prerequisite
1	ENG 1011	English I	3.0	
2	BDS 1201	History of the Emergence of Bangladesh	2.0	
3	CSE 1110	Introduction to Computer Systems	1.0	
4	MATH 1151	Fundamental Calculus	3.0	
Subtotal			9.0	

Trimester 2

Sl. No.	Course Code	Course Title	Credit Hr.	Prerequisite
1	ENG 1013	English II	3.0	ENG 1101
2	CSE 1111	Structured Programming Language	3.0	CSE 1110
3	CSE 1112	Structured Programming Language Laboratory	1.0	CSE 1110
4	CSE 2213	Discrete Mathematics	3.0	
Subtotal			10.0	

Trimester 3

Sl. No.	Course Code	Course Title	Credit Hr.	Prerequisite
1	MATH 2183	Calculus and Linear Algebra	3.0	MATH 1151
2	PHY 2105	Physics	3.0	
3	PHY 2106	Physics Laboratory	1.0	
4	CSE 2215	Data Structure and Algorithms I	3.0	CSE 1111
5	CSE 2216	Data Structure and Algorithms I Laboratory	1.0	CSE 1112
Subtotal			11.0	

Trimester 4

Sl. No.	Course Code	Course Title	Credit Hr.	Prerequisite
1	MATH 2201	Coordinate Geometry and Vector Analysis	3.0	MATH 1151
2	CSE 1325	Digital Logic Design	3.0	
3	CSE 1326	Digital Logic Design Laboratory	1.0	
4	CSE 1115	Object Oriented Programming	3.0	CSE 2215
5	CSE 1116	Object Oriented Programming Laboratory	1.0	CSE 2216
Subtotal			11.0	

Trimester 5

Sl. No.	Course Code	Course Title	Credit Hr.	Prerequisite
1	MATH 2205	Probability and Statistics	3.0	MATH 1151
2	SOC 2101	Society, Environment and Engineering Ethics	3.0	
3	CSE 2217	Data Structure and Algorithms II	3.0	CSE 2215
4	CSE 2218	Data Structure and Algorithms II Laboratory	1.0	CSE 2216
5	EEE 2113	Electrical Circuits	3.0	
Subtotal			13.0	

Trimester 6

Sl. No.	Course Code	Course Title	Credit Hr.	Prerequisite
1	CSE 3521	Database Management Systems	3.0	CSE 2215
2	CSE 3522	Database Management Systems Laboratory	1.0	CSE 2216
3	CSE 4165	Web Programming	3.0	CSE 1115, CSE 1116
4	EEE 2123	Electronics	3.0	EEE 2113
5	EEE 2124	Electronics Laboratory	1.0	
Subtotal			11.0	

Trimester 7

Sl. No.	Course Code	Course Title	Credit Hr.	Prerequisite
1	CSE 3313	Computer Architecture	3.0	CSE 1325
2	CSE 2118	Advanced Object Oriented Programming Laboratory	1.0	CSE 1116
3	BIO 3105	Biology for Engineers	3.0	
4	CSE 3411	System Analysis and Design	3.0	CSE 3521
5	CSE 3412	System Analysis and Design Laboratory	1.0	CSE 3522
Subtotal			11.0	

Trimester 8

Sl. No.	Course Code	Course Title	Credit Hr.	Prerequisite
1	CSE 4325	Microprocessors and Microcontrollers	3.0	CSE 3313
2	CSE 4326	Microprocessors and Microcontrollers Laboratory	1.0	EEE 2124
3	CSE 3421	Software Engineering	3.0	CSE 3411
4	CSE 3422	Software Engineering Laboratory	1.0	CSE 3412
5	CSE 3811	Artificial Intelligence	3.0	MATH 2205, CSE 2217

Course Sequence

6	CSE 3812	Artificial Intelligence Laboratory	1.0	MATH 2205, CSE 2218
Subtotal			12.0	

Trimester 9

Sl. No.	Course Code	Course Title	Credit Hr.	Prerequisite
1	CSE 2233	Theory of Computation	3.0	
2	PMG 4101	Project Management	3.0	CSE 3411
3	CSE 3711	Computer Networks	3.0	CSE 2217
4	CSE 3712	Computer Networks Laboratory	1.0	
5	CSE 4889	Machine Learning	3.0	CSE 3811, CSE 3812, MATH 2183
Subtotal			13.0	

Trimester 10

Sl. No.	Course Code	Course Title	Credit Hr.	Prerequisite
1	CSE 4000 A	Final Year Design Project - I	2.0	
2	CSE ****	Elective – I	3.0	
3	CSE 4509	Operating Systems	3.0	CSE 2217, CSE 3313
4	CSE 4510	Operating Systems Laboratory	1.0	CSE 2218
5	GED OPT1	General Education Optional-I	3.0	
Subtotal			12.0	

Trimester 11

Sl. No.	Course Code	Course Title	Credit Hr.	Prerequisite
1	CSE 4000 B	Final Year Design Project - II	2.0	CSE 4000 A
2	CSE ****	Elective – II	3.0	
3	CSE ****	Elective – III	3.0	
4	CSE 4531	Computer Security	3.0	CSE 3711, CSE 4509
5	GED OPT2	General Education Optional-II	3.0	
Subtotal			14.0	

Trimester 12

Sl. No.	Course Code	Course Title	Credit Hr.	Prerequisite
1	CSE 4000 C	Final Year Design Project - III	2.0	CSE 4000 A, CSE 4000 B

Course Sequence

2	CSE ****	Elective – IV	3.0	
3	CSE ****	Elective – V	3.0	
4	EEE 4261	Green Computing	3.0	
5	GED OPT3	General Education Optional-III	3.0	
Subtotal			14.0	

Credit hours distribution in twelve trimesters

Trimester	Theory Credits	Laboratory Credits	Total Credits
Trimester 1	8.0	1.0	9.0
Trimester 2	9.0	1.0	10.0
Trimester 3	9.0	2.0	11.0
Trimester 4	9.0	2.0	11.0
Trimester 5	12.0	1.0	13.0
Trimester 6	9.0	2.0	11.0
Trimester 7	9.0	2.0	11.0
Trimester 8	9.0	3.0	12.0
Trimester 9	12.0	1.0	13.0
Trimester 10	11.0	1.0	12.0
Trimester 11	14.0	0.0	14.0
Trimester 12	14.0	0.0	14.0
Total	125.0	16.0	141.0

Course Contents

Language Courses

ENG 1011: English I

Reading and Writing Cohesion, Skimming, Coherence, Scanning; Main ideas, Brainstorming and Taking notes, Comprehensions; Linking and Transitional words; Grammatical Knowledge: Parts of Speech, Punctuation, Subject-Verb Agreement, Preposition, Tense, Article, WH Questions, Paraphrasing; Summarizing; Creative Writing; Presentation

Speaking and Listening Speaking and Listening strategies; Pronunciation and Intonation; Vocabulary, Educated guess from content; Linking words and Fillers; Introduction to Drama; Performing Play; Art of Questioning; Famous Speeches; Listening Activities; How to make and present a brochure; Impromptu Speaking; Group Presentation

ENG 1013: English II

Prerequisite Course: ENG 1011

Writing Free Writing; Guided Writing: Paragraph writing with guidelines (based on hints, Wh questions); Process of Writing; Structure-based Paragraph Writing (types: Descriptive, Narrative and Process); Editing (Identification and correction of mistakes in Articles, Capitalization, Homonym, Fragment, Preposition, Pronoun, Punctuations, Run-on sentences, Faulty parallelism, Spelling, Subject-verb agreement, Tense); Application Writing; Email Writing; Steps of essay writing; Essay Writing in 5 paragraphs: (Cause and Effect essay, Compare and Contrast essay, Argumentative essay); Vocabulary: Sentence making practice on Academic word list (1-10)

Reading Practice on Reading Comprehensions

Speaking Public speaking; Argumentative Presentation

Listening Listening practice from various sources

General Education Courses

SOC 2101: Society, Environment and Engineering Ethics

Society: emergence of Sociology as moral lessons for society; Basic institutions in society, organization and institutions in society, Types of Society; Culture: basics of culture, elements of culture, cultural change, socialization, and social issues around us; Technology and society: interaction between technology and society; Engineering ethics: understanding ethics, engineering ethics; Moral reasoning and engineering as social experimentation; The engineers' concern for safety, professional responsibility; Employer authority; Rights of engineers; Global issues; Career choice and professional outlook; Ethical problems are like design problems; Genetically modified objects (GMO); Environment: environment and environmental issues– environmental degradation, waste management and renewable energy; Basic understanding of sustainable development, SDGs, climate change adaptation; Disability and Accessibility.

PMG 4101: Project Management

Prerequisite Course: CSE 3411

Triple Constraint in Project Management: Time, Scope and Cost; Process methodology, Requirement Collection, Plan, schedule a project including risk assessment with proper documentation and presentation. Cost Estimation, Optimization, and performance calculation, Change management, Quality improvement, Use of

Course Contents

Mod-ern tools in project planning, resource allocation and estimation.

BDS 1201: History of the Emergence of Bangladesh

Partition of Bengal (1947); Language Movement (1952); Movement for Autonomy; 6-point and 11-Point Programs; The 1970 Election; Speech on 7th of March 1971; Military Action, Genocide in the East Pakistan; The Liberation War; The Emergence of Bangladesh as a Sovereign Independent State in 1971; Constitution of Bangladesh and citizen rights; Culture: Cultural diffusion and change, Bengali culture and problems of society; social problems of Bangladesh; Social change: theories of social change; social change in Bangladesh; urbanization process and its impact on Bangladesh society.

ECO 4101: Economics

Definition of Economics; Economics and engineering; Principles of economics.

Micro-Economics: Introduction to various economic systems – capitalist, command and mixed economy; Fun-damental economic problems and the mechanism through which these problems are solved; Theory of demand and supply and their elasticities; Theory of consumer behavior; Cardinal and ordinal approaches of utility analysis; Price determination; Nature of an economic theory; Applicability of economic theories to the problems of devel-oping countries; Indifference curve techniques; Theory of production, production function, types of productivity; Rational region of production of an engineering firm; Concepts of market and market structure; Cost analysis and cost function; Small scale production and large scale production; Optimization; Theory of distribution; Use of derivative in economics: maximization and minimization of economic functions, relationship among total, marginal and average concepts.

Macro-Economics: Savings; investment, employment; national income analysis; Inflation; Monetary policy; Fiscal policy and trade policy with reference to Bangladesh; Economics of development and planning.

SOC 4101: Introduction to Sociology

Concept and theory: major schools of sociology - functionalism, critical theory, gender, interactionism and post-modernism; Sociology of communications: the impacts of contemporary media institutions and communications technologies on the social construction of knowledge and the construction of socially significant identities and ide-ologies; Society: discussion on key concepts of society, social institutions, social structure and stratification, religion and so on; Sociology of development: technology, gender, business, globalization, and how do we formulate rea-sonable expectations? Global and social issues; Social research: importance of research, research methods and techniques.

ACT 211: Financial and Managerial Accounting

Financial Accounting: Objectives and importance of accounting; Accounting as an information system; Comput-erized system and applications in accounting. Recording system: double entry mechanism; accounts and their classification; Accounting equation; Accounting cycle: journal, ledger, trial balance; Preparation of financial state-ments considering adjusting and closing entries; Accounting concepts (principles) and conventions.

Financial statement analysis and interpretation: ratio analysis.

Cost and Management Accounting: Cost concepts and classification; Overhead cost: meaning and classification; Distribution of overhead cost; Overhead recovery method/rate; Job order costing: preparation of job cost sheet and quotation price; Inventory valuation: absorption costing and marginal/variable costing techniques; Cost-Volume-Profit analysis: meaning breakeven analysis, contribution margin approach, sensitivity analysis.

Short-term investment decisions: relevant and differential cost analysis. Long-term investment decisions: capital budgeting, various techniques of evaluation of capital investments.

IPE 3401: Industrial and Operational Management

Introduction, evolution, management function, organization and environment.

Organization: Theory and structure; Coordination; Span of control; Authority delegation; Groups; Committee and task force; Manpower planning.

Personnel Management: Scope; Importance; Need hierarchy; Motivation; Job redesign; Leadership; Participative management; Training; Performance appraisal; Wages and incentives; Informal groups; Organizational change and conflict.

Cost and Financial Management; Elements of costs of products depreciation; Break-even analysis; Investment analysis; Benefit cost analysis.

Management Accounting: Cost planning and control; Budget and budgetary control; Development planning process.

Marketing Management: Concepts; Strategy; Sales promotion; Patent laws.

Technology Management: Management of innovation and changes; Technology life cycle; Case studies.

GED 4003: Technology Entrepreneurship

Entrepreneurial and Technological Ecosystems: Entrepreneurship, Intrapreneurship, Entrepreneurial Process; Developing and Validating Solutions and MVPs: Entrepreneur's Vision and Concept, Entrepreneurial Context, a New Venture, Technological Feasibility, Market Feasibility; Team Management: : Principles of Human Resources and Talent, Teamwork, Project Group Self-evaluation Activity; Business Model Design and Innovation: Business Plans, Start-up Strategies, Business Idea Evaluation, Creativity and Improvisation; Sales and Marketing for Technology Startups: Customer Discovery Process, Conducting Market Research, Market Analysis, Marketing Strategy; IP and Legal Issues for Technology Focused Firms: Patents, Copyright, Trademark, Trade Secret, Leverage IP in Strategies with investors, partners, competitors; Initial Public Offerings(IPO), Public Company and Industry Metrics; Financing and Pitching New Ventures: Investors and Venture Capital, Principles of Venture Finance, Principles of Venture Finance, Risk Reduction, Ecosystem of Investors and Partners; Social Entrepreneurship: Professional Ethics and Personal Integrity, Social Entrepreneurship Drivers - scalability, sustainability, opportunity, accountability.

PSY 2101: Psychology

The objective of this course is to provide knowledge about the basic concepts and principles of psychology pertaining to real-life problems. The course will familiarize students with the fundamental processes that occur within organism-biological basis of behavior, perception, motivation, emotion, learning, memory and forgetting and also to the social perspective-social perception and social forces that act upon the individual.

SOC 4301: Introduction to Anthropology

Scope and History of anthropology; Anthropology and Colonialism; Different branches of anthropology; Anthropology and other Disciplines; Theory and methods in anthropology; Core Concepts: Culture, Marriage and Family, Kinship, Descent and Social Structure; Economic Systems, Political Systems; Applying Anthropology: Applied Anthropology; Recent Trends in Anthropology: Changes and Direction; Research methods to study human behavior;

Computational Methods in Anthropology.

FLN 1101: Introduction to a Foreign Language

Elementary proficiency in oral expression, listening comprehension, reading and writing in a foreign language; cul-tural understanding on topics related to daily activities and personal environment; basic grammatical knowledge and vocabulary for common use; ability to communicate in the most predictable and common situations using high-frequency words, expressions, and phrases.

SWL 4101: Software and Law

Introduction: Why law is necessary, the laws relevant to software and its use, taking a global perspective. Intel-lectual property (IP) rights: intellectual-property law, copyright, patents, marks and brands; common and public goods. The issue of who owns software and digital content how that ownership can be protected using instruments like copyright and patents. Contracts: how contracts are used to formalize the relationship between purchasers and suppliers of software. Responsibilities to employees and the public: the laws that apply to information technology and software to ensure that organizations meet their responsibility of care towards employees, customers, and the general public.

BHV 2101: Introduction to Behavioral Science

Basic scientific theoretical perspectives, epistemological views and theories of the behavioral sciences; central con-cepts, theories, models and research results in the behavioral sciences; integration of educational, psychological, and sociological theories in the behavioral science field; diversity perspectives based on class, gender and ethnicity in relation to behavioral science theories; relevant search tools and critically review search results.

MKT 4313 : Digital Marketing

Digital Marketing Essentials: The Digital Marketing Environment; The Digital Consumer; Content Marketing; Digital Marketing Tools and Channels: Social Media Marketing; Augmented, Virtual and Mixed Reality, Search Engine Optimization (SEO); Search Engine Marketing (SEM); Digital Marketing Strategy and Planning: Digital Marketing Strategy and Objectives; Digital Marketing Management: Integrating, Improving and Transforming Digital Marketing; Further issues in legal and social Sensitivity.

BDS 2201: Bangladesh Studies

Ancient Bengal: Sasanka, Rise of the Palas, the Senas; Early Medieval Bengal; Coming of the Muslims; The In-dependent sultanate of Bengal: Ilyas Shahi and Hossein Shahi Bengal; Late medieval Bengal: The Establishment of Mughal Rule in Bengal; Bara Bhuiyans: Subedars and Nawabs; The European Style in Bengal Architecture; British rule in Bengal; Battles of Plassey and Buzas; The Dual government; permanent settlement (1793); Nine-teenth century Bengali renaissance: social and religious reforms, Raja Rammohan Roy, Ishwar Chandra Vidyasagar, Titu Meer; Partition of Bengal (1905); Language Movement (1952); Movement for Autonomy; 6-point and 11-Point Programs; The 1970 Election-Military Action, Genocide in the East Pakistan; The Liberation War; The Emergence

of Bangladesh as a Sovereign Independent State in 1971; Culture: Cultural diffusion and change, Bengali culture and problems of society; social problems of Bangladesh; Social change: theories of social change; social change in Bangladesh; urbanization process and its impact on Bangladesh society.

BAN 2501: Bangla

(অ) বাংলা সাহিত্য

ক। নির্বাচিত প্রবন্ধ : (যে কোনো ৩টি)

(১) হরপ্রসাদ শাস্ত্রী : তৈল, (২) বঙ্কিমচন্দ্র চট্টোপাধ্যায় : বাঙালা ভাষা (৩) রবীন্দ্রনাথ ঠাকুর : সভ্যতার সংকট, (৪) প্রমথ চৌধুরী : বীরবলের হালখাতা (যে কোনো ১টি প্রবন্ধ) (৫) মোতাহের হোসেন চৌধুরী : শিক্ষা ও মনুষ্যত্ব (৬) অন্যান্য প্রবন্ধ (সহায়ক গ্রন্থ হতে নির্বাচিত)

খ। নির্বাচিত গল্প : (যে কোনো ৩টি)

(১) রবীন্দ্রনাথ ঠাকুর : পোস্টমাস্টার / স্ত্রীর পত্র/ একরাত্রি (২) বনফুল : নিমগাছ (৩) বিভূতিভূষণ বন্দ্যোপাধ্যায় : পুঁই মাচা (৪) বেগম রোকেয়া সাখাওয়াত হোসেন : অবোরোধবাসিনী (৫) সৈয়দ ওয়ালীউল্লাহ : নয়নচারা (৬) অন্যান্য গল্প (সহায়ক গ্রন্থ হতে নির্বাচিত)

গ। নির্বাচিত কবিতা : (যে কোনো ৩টি)

(১) রবীন্দ্রনাথ ঠাকুর : নির্ঝরের স্বপ্নভঙ্গ, (২) কাজী নজরুল ইসলাম : আজ সৃষ্টি সুখের উল্লাসে (৩) জীবনানন্দ দাশ : বনলতা সেন (৪) শামসুর রাহমান : তোমাকে পাওয়ার জন্য হে স্বাধীনতা (৫) নির্মলেন্দু গুণ : ছলিয়া (প্রেমাংশুর রক্ত চাই) (৬) অন্যান্য কবিতা (সহায়ক গ্রন্থ হতে নির্বাচিত)

ঘ। উপন্যাস (যে কোনো ১টি)

বিভূতিভূষণ বন্দ্যোপাধ্যায় : আরণ্যক, অদ্বৈত মল্লবর্মণ : তিতাস একটি নদীর নাম, মানিক বন্দ্যোপাধ্যায় : দিবারাত্রির কাব্য

(আ) প্রায়োগিক বাংলা

(ক) বাংলা ভাষায় লিখন-দক্ষতা

(১) বাংলা ধ্বনিতত্ত্ব (ধ্বনি, বর্ণ, ধ্বনি পরিবর্তন, যুক্তবর্ণ), (২) বাংলা বানান : বাংলা একাডেমির বাংলা বানানের নিয়ম, শব্দের অপপ্রয়োগ, শব্দের বানান ও অশুদ্ধি, (৩) বাক্যের শুদ্ধি-অশুদ্ধি : বাক্যের গঠনগত শুদ্ধি-অশুদ্ধি, বিরাম চিহ্ন, (৪) বাংলা লিখন কৌশল : রেজুলেশন লিখন, অনুষ্ঠান সংগলন পাণ্ডুলিপি প্রস্তুত, বিজ্ঞাপন লিখন, প্রুফ সংশোধন।

(খ) বাংলা ভাষায় শ্রবণ ও কথন-দক্ষতা

(১) বাংলা উচ্চারণের নিয়ম : স্বরবর্ণ ও ব্যঞ্জনবর্ণের উচ্চারণের স্থান, উচ্চারণরীতি, (২) বাংলা উচ্চারণ-সূত্র ও তার প্রয়োগ সহায়ক গ্রন্থ :

(১) প্রবন্ধ সংগ্রহ, ঢাকা বিশ্ববিদ্যালয় প্রকাশনা সংস্থা, (২) গল্প সংগ্রহ, ঢাকা বিশ্ববিদ্যালয় প্রকাশনা সংস্থা, (৩) কবিতা সংগ্রহ, ঢাকা বিশ্ববিদ্যালয় প্রকাশনা সংস্থা, (৪) বাংলা ভাষার ব্যাকরণ, মাহবুবুল আলম

URC 1101: Life Skills for Success

The course is intended for fresh entrants at the first trimester who need to be oriented and adapted to university survival skills, as well as achieving soft skills for success as a responsible citizen in the society. Complementary to this core object, students need to be motivated and inspired to study attentively with a sense of integrity and ethical orientation. In addition, this course will create students' awareness to build a successful career as well as becoming a successful individual in the society. The course will cover lectures on rules and regulations of the university, the importance of student life, contribution of family, building professional ethics and personal integrity, time management, study skills, etiquettes and manners, social responsibility including environmental concerns, effective communication, dealing with health and psychological issues, etc. The course is expected to take care of this broad gamut of soft skills that would immensely inspire towards developing a quality person.

Basic Sciences Courses

PHY 2105: Physics

Waves and Oscillations Periodic motion: periodic waves, elastic restoring force, simple harmonic motion (SHM), differential equation of SHM and its solutions, examples of SHM, energy calculation of SHM, time period, velocity, acceleration, frequency calculation with graph, Lissajou's figure design, spring mass system and torsional pendulum, DHM, characteristic graph, differential equations for spring mass system with damping mechanism and RLC circuit-series and parallel analysis, resonant frequency, reactance, impedance, FHM; Mechanical Waves; Vibrating bodies and acoustic phenomena: progressive wave and its differential equation, EM wave, group velocity, phase velocity, standing waves, node and antinode; The Doppler effects, application of acoustic Phenomena.

Electricity magnetism Electrostatic Force and Electric Field; Concept of charge, Coulomb's law, concept of electric field and its calculation, electric dipole; Gauss's law in electrostatic and its application, electric field due to dipole, torque on a dipole in uniform e-field, electric flux, flux density, Gauss's law and Coulomb's law; Electric Potential: electric potential and its calculation, electric potential energy, relationship between field and potential, potential due to a point charge, dipole, continuous charge distribution, electric field calculation from electric potential, equipotential surface, potential gradient; Capacitance and Dielectric : capacitors, capacitors in series and parallel, energy of charged capacitors, electrical energy density in terms of electric field, electron volt, dielectric media, polarization vector and displacement vector, Laplace's and Poission's equations, capacitor with a dielectric material, Gauss's law with dielectric; Current, Resistance and Electromotive Force: current and current density, resistance and resistivity, Ohm's law, EMF, power, resistance in series and parallel, Kirchhoff's Rules, RC circuit; Magnetic Field: magnetic field, magnetic flux and flux density, Lorentz force, Gauss's law for magnetism, motion of a charged particles in magnetic field : Hall effect; Magnetic field intensity, magnetic dipole moment, Biot-Savart Law, Ampere's law and its applications; Magnetic properties of material, magnetization, hysteresis; Inductions and Inductance: induced emf and Faraday's law of induction; Lenz's law; Mutual inductance ; Self inductance; Energy in an inductor; Inductance in series, in parallel, and their combination, MMF, leakage and fringing flux, Transformers.

Quantum Physics Quantum theory: quantum theory of radiation, energy of photons, photo-electric Effect, work function, threshold frequency, threshold voltage, Compton effect, X-rays production, properties and applica-tion, Bragg Diffraction, De Broglie wave length, Heisenberg' s Uncertainty Principle, correspondence principle, pair production, pair annihilation; Schrodinger equation: wave function, Schrodinger equation-time dependent and time independent form, expectation value, quantum operator, tunneling effect, quantum numbers, energy of trapped electron, quantum dots and corrals, quantization of Bohr orbital energy.

PHY 2106: Physics Laboratory

Experiments based on PHY 2105

BIO 3105: Biology for Engineers

Introduction; The Basics of Life: Chemistry; Organic Molecules: The Molecules of Life; Cell Structure and Function; Enzymes, Coenzymes, and Energy; Biochemical Pathways: Cellular Respiration, Photosynthesis; DNA and RNA: The Molecular Basis of Heredity; Cell Division; Patterns of Inheritance; Applications of Biotechnology; Diversity within Species and Population Genetics; Evolution and Natural Selection; The Formation of Species and Evolution-ary Change; Ecosystem Dynamics: The Flow of Energy and Matter; Community Interactions; Population Ecology; Evolutionary and Ecological Aspects of Behavior; The Origin of Life and Evolution of Cells; The Classification and Evolution of Organisms; The Nature of Microorganisms; The Plant Kingdom; The Animal Kingdom; Mate-rials Exchange in the Body; Nutrition: Food and Diet; The Body's Control Mechanisms and Immunity; Human Reproduction, Sex, and Sexuality.

Mathematics Courses

MATH 1151: Fundamental Calculus

Function, Domain and Range of a Function. Translation and reflection of a function. Even and Odd functions, Inverse functions, One to One and many to one function. Limit, continuity and differentiability, Tangent line, Differentiation of different types of functions. An overview of area problem, Newton's anti-derivative method in finding area, Indefinite integral, fundamental theorem of calculus, Definite integral, Area between two curves. Different types of Integration (Principles of Integral evaluation, Integration by parts, Trigonometric Substitution).

MATH 2183: Calculus and Linear Algebra

Prerequisite Course: MATH 1151

Calculus: Analysis of Function I: Slope and Concavity, Analysis of function II: Relative Extrema and Polyno-mials, Partial Derivatives, The Chain Rule.

Differential Equation: Solution of the differential equations of 1st and 2nd order.

Linear Algebra: Solution of different types of system of linear equations. Operations of matrix algebra, trans-position, inversion, rank of matrices. Solution of system of equations by matrix method. Eigen values and Eigen vectors.

MATH 2201: Coordinate Geometry and Vector Analysis

Prerequisite Course: MATH 1151

Conic sections, rotation of axes, Rectangular co-ordinate in 3-space, cross and dot product of vectors, parametric equation of straight lines, plane in 3-space. Cylindrical and spherical coordinate systems, integrals of multi-variable functions (double and triple integrals including polar coordinates). Gradient of scalar fields, divergence and curl of vector fields. Line integrals, conservative vector field and Green's theorem, surface integral, flux, divergence theorem, Stokes' theorem.

MATH 2205: Probability and Statistics

Prerequisite Course: MATH 1151

Frequency distribution; Mean, median, mode and other measures of central tendency; Standard deviation and other measures of dispersion; Moments, skewness and kurtosis, correlation and regression analysis; Elementary probability theory and discontinuous probability distribution, e.g., binomial, Poisson and negative binomial; Continuous probability distributions, e.g. normal and exponential; Characteristics of distributions; Elementary sampling theory; Estimation of parameter, Hypothesis testing.

Other Engineering Courses

EEE 2113: Electrical Circuits

Fundamental electrical concepts and measuring units, DC voltages, current, resistance and power, laws of electrical circuits and methods of network analysis, principles of DC measuring apparatus, laws of magnetic fields and methods of solving simple magnetic circuits; Alternating current: instantaneous and RMS current, voltage and power, average power combinations of R, L & C circuits, phasor, representation of sinusoidal quantities.

EEE 2123: Electronics

Prerequisite Course: EEE 2113

Semiconductor diode: materials, energy band, n-type and p-type materials, p-n junction diode, ideal vs practical diode, zener diode, light emitting diode; Diode applications: load-line Analysis, series-parallel dc circuits, AND/OR logic gates, full-wave and half-wave rectification, clipper and clamper circuits; Bipolar junction transistors: device structure and physical operation, current-voltage characteristics, BJT Circuits at DC, BJT as an amplifier and as a switch; MOS field-effect transistors (MOSFETs): device structure and physical operation, current-voltage characteristics, MOSFET circuits at DC, MOSFET as an amplifier and as a switch; CMOS combinational logic circuit design.

EEE 2124: Electronics Laboratory

Laboratory work based on EEE 2123.

EEE 4261: Green Computing

Cloud computing: Definition, Concept, service model and their clarification, deployment model, security and privacy; Edge Computing: Definition, Concept, Advantages and challenges; Tele-computing: Definition, advantages and challenges; Power and energy management: IEEE rules and codes in power and energy management, Microsoft, IBM and others definition in energy management; E-waste: Definition of e-waste and its recycle process. Cost benefit analysis of e-waste recycle. And environmental impact analysis of e-waste.

Core Courses

CSE 1110: Introduction to Computer Systems

Introduction to computations; Early history of computing devices; Computers; Major components of a computer; Hardware: processor, memory, I/O devices; Software: Operating system, application software; Basic architecture of a computer; Basic Information Technology; The Internet; Number system: binary, octal, hexadecimal, binary arithmetic; Basic programming concepts; Program development stages: flow charts; Programming constructs: data types, operators, expressions, statements, control statements, functions, array.

CSE 1111: Structured Programming Language

Prerequisite Course: CSE 1110

Basic understanding of problem solving; Structured programming language: data types, operators, expressions, control structures (If-else, Switch-case, Loop); Functions and program structure: parameter passing conventions, scope rules and storage classes, recursion; Header files; Pointers and arrays; Strings; Multidimensional array; User defined data types: structures, unions, enumerations; Input and Output: standard input and output, formatted input and output, file access; Variable length argument list; Command line parameters; Error Handling; Graphics; Linking; Library functions.

CSE 1112: Structured Programming Language Laboratory

Prerequisite Course: CSE 1110

Laboratory work based on CSE 1111 with a project work.

CSE 1115: Object Oriented Programming

Prerequisite Course: CSE 2215

Philosophy of Object Oriented Programming (OOP); Advantages of OOP over structured programming; Abstraction and Encapsulation, classes and objects, access specifiers, static and non-static members; Constructors, destructors and copy constructors; Array of objects, object pointers, and object references; Inheritance: single and multiple inheritance; Polymorphism:overloading, abstract classes, virtual functions and overriding; Exceptions; Object Oriented I/O; Template functions and classes; Multi-threaded Programming.

CSE 1116: Object Oriented Programming Laboratory

Prerequisite Course: CSE 2216

Laboratory work based on CSE 1115.

CSE 2118: Advanced Object Oriented Programming Laboratory

Prerequisite Course: CSE 1116

Laboratory work based on advanced topics in Object Oriented Programming with a project work.

CSE 4165: Web Programming

Prerequisite Course: CSE 1115, CSE 1116

Web architecture and HTTP: History and architecture of the World Wide Web, overview of the Hyper Text Transfer Protocol, other related protocols; Hyper Text Markup Language: The concept of markup, overview of HTML (table, form, frame, window, link etc.); Cascading Style Sheets: Overview of CSS (selectors, different CSS properties and values); Client side scripting: Variables, data types, control structure, functions, Document Object Model (DOM), event handlers, properties, methods, cookies; Server side scripting: Concepts, variables, data types, control structure, functions, objects, regular expressions, mails, cookies, sessions and a related web framework; Database: Content generation, data exchange; Layered or Multi-tier Architecture for Web Applications; MVC; Content Management System.

CSE 1325: Digital Logic Design

Number systems: Introduction, digital number systems, arithmetic operations; Function minimization techniques: Boolean algebra (identities, functions and manipulation), Canonical and standard forms, minimization techniques; Combinational logic circuits design procedure; Combinational and Arithmetic functions: Arithmetic (adders) and other popular (encoders, decoders, multiplexers, demultiplexers) modules; Sequential circuits and Registers: Sequential logic design procedure, state diagrams, state table, input and output equations, latches, flip-flops, race around problems, design problems, registers, register transfers, counters and their applications.

CSE 1326: Digital Logic Design Laboratory

Laboratory work based on CSE 1325.

CSE 3313: Computer Architecture

Prerequisite Course: CSE 1325

Information representation; Measuring performance; Instructions and data access methods: operations and operands of computer hardware, representing instruction, addressing styles; Arithmetic Logic Unit (ALU) operations, floating point operations, designing ALU; Processor design: datapath - single cycle and multicycle implementations; Control Unit design - hardwired and microprogrammed; Pipeline: pipelined datapath and control, hazards; Exceptions; Memory organization: Cache, Concepts of DMA and Interrupts; Buses: overview of computer BUS standards; Multiprocessors: types of multiprocessors, performance, single bus multiprocessors, multiprocessors connected by network, clusters.

CSE 4325: Microprocessors and Microcontrollers

Prerequisite Course: CSE 3313

Introduction to 16-bit and 32-bit microprocessors: architecture, addressing modes, instruction set (e.g. x86), interrupts, multitasking and virtual memory, paging, cache memory; Interfacing: programmable peripheral interface, direct memory access (DMA), keyboard and display interface, memory chips (e.g. ROM, RAM), clock generator, bus arbiter; Architecture of modern microprocessors: multi processors vs multi-core architecture, hyperthreading technology, turbo boost technology; Introduction to micro-controllers (e.g. ATmega32): architecture, digital interfacing: LEDs, switches, sensors and motors, analog interfacing: introduction to the analog-to-digital converter (ADC) module, serial communication protocols (e.g. SPI, I2C, CANBUS) for embedded system.

CSE 4326: Microprocessors and Microcontrollers Laboratory

Prerequisite Course: EEE 2124

Students will design simple systems using the principles learned in CSE 4325. An introduction to assembly language will be included in this course at the beginning.

CSE 2213: Discrete Mathematics

Set theory: sets, relations, functions; Mathematical Logic: propositional calculus and predicate calculus; Mathematical reasoning and proof techniques; Counting: permutations, combinations, Discrete Probability principles of inclusion and exclusion; Recurrence relations; Graph Theory: graphs, paths, and trees.

CSE 2215: Data Structure and Algorithms - I

Prerequisite Course: CSI 1111

Internal data representation; Abstract data types; Introduction to algorithms; Asymptotic analysis: growth of functions, O , Ω and Θ notations; Correctness proof and techniques for analysis of algorithms; Master Theorem; Elementary data structures: arrays, linked lists, stacks, queues, trees and tree traversals, graphs and graph representations, heaps, binary search trees; Graph Traversals: DFS, BFS, Applications of DFS and BFS; Sorting: heap sort, merge sort, quick sort, linear-time sorting; Data structures for set operations.

CSE 2216: Data Structure and Algorithms - I Laboratory

Prerequisite Course: CSE 1112

Laboratory work based on CSE 2215.

CSE 2217: Data Structure and Algorithms - II

Prerequisite Course: CSE 2215

Methods for the design of efficient algorithms: divide and conquer, greedy methods, dynamic programming; Graph algorithms: MST algorithms, shortest path algorithms, maximum flow and maximum bipartite matching; Advanced data Structures: balanced binary search trees (AVL trees, red-black trees, splay trees), skip lists, advanced heaps (Fibonacci heaps, binomial heaps); Hashing; String matching algorithms; NP-completeness; NP-hard and NP-complete problems; Coping with hardness: backtracking, branch and bound, approximation algorithms.

CSE 2218: Data Structure and Algorithms - II Laboratory

Prerequisite Course: CSE 2216

Laboratory work based on CSE 2217.

CSE 2233: Theory of Computation

Finite Automata: Deterministic finite automata, Non-deterministic finite automata, equivalence and conversion of deterministic and non-deterministic finite automata, pushdown automata. Context free language, context free grammar. Turing machines: basic machines, configuration, computing with Turing machine, combining Turing machines.

CSE 3411: System Analysis and Design

Prerequisite Course: CSE 3521

System Study: concept of system and system study, system organogram, system development life cycle, different types of system, skills of system analyst; Information Gathering: types of information, sources of information, information gathering tools and their competitive analysis; Guidelines to conduct information gathering tools

Feasibility Study: statement of constraints, types of feasibility for IT products, determining best candidate system, SWOT analysis, cash flow and NPV analysis, feasibility Report; System Design: structured and object oriented design using UML; DFD, use case, sequence diagram, state diagram, class diagram, etc using UML tools; Effective designing of input, output and UI; Software Requirement Specifications (SRS); Project deployment: study on project management and tools; Scheduling by Gantt chart, PERT/CPM method, etc; System security, risk management, data migration, training, art of negotiation, etc.

CSE 3412: System Analysis and Design Laboratory

Prerequisite Course: CSE 3522

Laboratory work based on CSE 3411.

CSE 3421: Software Engineering

Prerequisite Course: CSE 3411

Basic Concepts: software, software engineering, recent trends and challenges; Process Models: waterfall, incremental, iterative; Requirements Engineering: software requirements specification, system requirements specification, stakeholder requirements specification; Architecture: monolithic architecture, service-oriented architecture, micro-service architecture, model-view-controller pattern and variants, system design; Services Computing: application programming interface, web services, cloud services, representational state transfer, JavaScript object notation, simple object access protocol; User Interface Design: web and mobile platform, wireframe model, methods and tools; Software Testing: manual and automated test, black box and white box test, unit test, integration test, regression test, acceptance test, non-functional test, test planning, test documentation; Version Control and Repository: version numbering, version control software, code repository systems; Documentation: requirements, architecture, technical, end user, marketing; Legal and Ethical Aspects: terms and conditions, end-user license agreement, software engineering code of ethics, privacy engineering; Business Case Study: case study on local and international popular software products.

CSE 3422: Software Engineering Laboratory

Prerequisite Course: CSE 3412

Laboratory work based on CSE 3421.

CSE 4531: Computer Security

Prerequisite Course: CSE 3711, CSE 4509

Fundamental concepts: confidentiality, integrity and availability, assurance, authenticity and anonymity; threats and attacks, security principles; Encryption, symmetric and asymmetric key encryption; Security: OS access control, Web and mobile application security, software security, hardware security, memory protection, data base security; Security Attacks: malware, DDos, Trojan and backdoors, buffer overflow, social engineering.

CSE 3521: Database Management Systems

Prerequisite Course: CSE 2215

Concepts of database systems; Data Models: Entity-Relationship model, Relational model; Query Languages: SQL, Relational algebra, Constraints, View; Security and Integrity Management; Functional dependencies and normalization; Indexing: primary and secondary indexes, B+ trees; Hashing: Static and Dynamic hashing, Collision Problem in Hashing; Transaction management; Recovery: RAID Different levels; File storage

Course Contents

management.

CSE 3522: Database Management Systems Laboratory

Prerequisite Course: CSE 2216

Laboratory work based on CSE 3521. A project work will be included.

CSE 4509: Operating Systems

Prerequisite Course: CSE 2217, CSE 3313

Operating system: its role in computer systems; multitasking, multiuser, multiprocessing OS; Operating system structures; Process: process concept and scheduling, inter-process communication, communication in client-server systems; CPU scheduling: scheduling criteria and algorithms, thread scheduling, multiple-processor scheduling; Process synchronization: critical-section problem, semaphores, monitors; Deadlock: resource allocation and dead-lock, deadlock detection, prevention and recovery; Memory management: swapping, paging, segmentation, virtual memory; File Systems: files, directories, security, protection; Case study of some operating systems.

CSE 4510: Operating Systems Laboratory

Prerequisite Course: CSE 2218

Laboratory work based on CSE 4509.

CSE 3711: Computer Networks

Prerequisite Course: CSE 2217

Introduction to Computer Networks; Network Edge, Network Core ; Layering architecture: TCP/IP and OSI Reference Models; Circuit Switching and Packet Switching; Hubs, Routers, and Switches; Application layer services: Web, HTTP, FTP, SMTP, DNS architecture; Introduction to transport layer: UDP,TCP; Principles of Reliable data transfer; TCP Congestion and Flow control; Routing and forwarding, DHCP, NAT, Fragmentation; Routing algorithms; Autonomous Systems; Link layer services; MAC Protocols; Link layer addressing; Ethernet; ARP; Wireless links and network characteristics; Wi-Fi: IEEE 802.11 Wireless LANs.

CSE 3712: Computer Networks Laboratory

Laboratory work based on CSE 3711.

CSE 3811: Artificial Intelligence

Prerequisite Course: MATH 2205, CSE 2217

Survey and concepts in Artificial Intelligence; Problem solving agents; Uninformed and Informed search tech-niques; Local Search Techniques; Game playing; Constraint Satisfaction Problems; Bayesian learning; Supervised Learning: Classification, Perceptrons; Stationary processes and Markov assumptions; Hidden Markov Models; Hu-man Aware AI Systems.

CSE 3812: Artificial Intelligence Laboratory

Prerequisite Course: MATH 2205, CSE 2218

Laboratory work based on CSE 3811.

CSE 4889: Machine Learning

Prerequisite Course: CSE 3811, CSE 3812, MATH 2183

Introduction to Machine Learning; Regression analysis: linear regression; Classification techniques: classification trees, support vector machines; Statistical performance evaluation: bias-variance tradeoff; VC dimension; Reinforcement Learning; Neural networks; EM Algorithm; Unsupervised Learning: k-means clustering; Principal component analysis; Deep Learning; Practical applications of machine learning.

Elective Courses

CSE 4121: Security Monitoring

Prerequisite Courses: CSE 4531

Fundamental principles of monitoring networked systems; collection and analysis of network telemetry; selection of network-centric security analytics techniques; identification of critical indicators of compromise and behavioral attributes; development of derived features from packet captures, flow data, and logs; discovering malicious patterns in traffic datasets; server-side compromise, client-side compromise, proxies and checksums, intrusion detection and anomaly-detection methodologies including signature-based, behavior-based, and machine-learning-based approaches; selecting appropriate monitoring and detection mechanisms for different threat models.

CSE 4759: Wireless and Cellular Communication

Prerequisite Course: CSE 3715, CSE 3711

Cellular concepts: frequency reuse, handoff strategies, interference and system capacity, grade of service, improving capacity and coverage, call blocking probability; Propagation effects: outdoor propagation models, indoor propagation models, power control, Doppler's effect, small and large scale fades; Wireless LAN Technology; IEEE 802.11: standard, protocol architecture, physical layer and media access control; Mobile IP; Wireless Application Protocol; IEEE 802.16 Broadband Wireless Access; Brief review of 2nd and 3rd generation wireless: GSM, GPRS, CDMA; LTE, LTE-Advanced, and 5G. Vehicular wireless networks, white spaces, IEEE 802.22 regional area networks, Bluetooth and Bluetooth Smart, wireless personal area networks, wireless protocols for Internet of Things, ZigBee.

CSE 4123 - Software and Application Security

Prerequisite Courses: CSE 3421, CSE 4531

Software and security concepts, software security problems, threats to security, software security metrics, software security estimation, secure software architecture, secure software development process, software security testing, case study on implementing software security.

CSE 4125: Ethical Hacking and Network Defense

Prerequisite Courses: CSE 4531: Computer Security

Ethical hacking concepts, categories of penetration test, information gathering techniques, target enumeration and port scanning techniques, vulnerability assessment, network sniffing, exploitation, post exploitation, hacking web servers, web application vulnerabilities, web based password cracking techniques, hacking wireless

networks; concepts of network security, fundamentals of firewalls, firewall practical applications, intrusion detection system, virtual private networks, operating system hardening, defending against virus attacks, trojan horses, spywares and adwares, security policies, assessing system security, security standards.

CSE 4127: Cloud Security

Prerequisite Courses: CSE 4531, CSE 4509

Principles and concepts, data asset management and protection, cloud asset management and protection, identity and access management, vulnerability management, network security; detecting, responding to, and recovering from security incidents; examples of cloud service providers.

CSE 4129: Digital Forensics

Prerequisite Courses: CSE 4531

Foundations, language of computer crime investigation, digital evidence in the courtroom, worldwide perspectives of cyber crime law, conducting digital investigations, handling a digital crime scene, investigative reconstruction with digital evidence; digital evidence on windows, linux and mobile devices; network forensics.

CSE 4131: Internet of Things Security

Prerequisite Courses: CSE 4531

Attack, defense and network robustness of Internet of Things, privacy preservation, trust and authentication in Internet of Things, computational security for Internet of Things, security protocols, social awareness, security and impact of Internet of Things on mobile networks.

CSE 3715: Data Communication

Introduction of layered network architecture; Introduction of data communication: physical point to point communication, signal, signal representation and processing, signal to noise ratio; Framing techniques; Frequency response of signals: Fourier integrals, Fourier transforms, time domain and frequency domain concept; representation of noise; Introduction to information theory: entropy, information capacity; Modulation and demodulation: amplitude modulation, frequency and phase Modulation; From analog to digital communication: sampling, Nyquist theorem, quantization, digitization of analog signals; Line coding; Techniques of modulation: pulse modulation, pulse amplitude modulation, pulse width modulation, pulse position modulation, pulse code modulation; Multi-plexing techniques: time division multiplexing, frequency division multiplexing techniques.

4793: Advanced Network Services and Management

Prerequisite Course: CSE 3711

Application specific protocols: domain name services, electronics mail; World Wide Web and Web caching; Network Management (SNMP), error Reporting Mechanism (ICMP), socket Interfaces, file transfer and remote file access; Multimedia application: RTP, session control; Intra- and Inter-AS routing: IGP, EGP, BGP; Network security: cryptography, firewalls, access control lists (ACLs); VPN, IPsec, IPv6.

CSE 4783: Cryptography

Prerequisite Course: CSE 4531

Cryptography, history of cryptography; Perfect ciphers, Stream ciphers, attacks on stream ciphers, block ciphers, how to use block ciphers with one time key and many time key; Symmetric encryption, , DES, TDES, AES, Feistel block structure; Asymmetric key: public key protocols, basic key exchange, RSA (cryptosystem); Quantum
-cryptography, one time pad exchange using qbits; Message integrity (MAC), HMAC, Secure hash functions. Digital signatures.

CSE 4777: Network Security

Prerequisite Course: CSE 4531

Introduction to computer security, CIA TRIAD, Threats and Attacks, Passive and Active attacks and examples of passive as well as active attacks, security mechanisms, network security model; Hashing, Cryptography, Intro-duction to Symmetric key and Asymmetric key encryption; One way authentication protocols, Needham Schroeder protocol, Needham-Schroeder Symmetric key protocol Anomaly in Needham Schroeder Symmetric key protocol, Needham-Schroeder Asymmetric key protocol (Kerberos); IP Sec, Intrusion Detection System (IDS) (Firewall), TLS, HTTPS, TELNET, SSH, Wire-shark; Wireless network security: WEP, WPA, WPA2; Secure Hash Algo-rithm (SHA), Digital Signature Standard (DSS); Advanced network security topics.

CSE 4943: Web Application Security

Client-side (browser) security: vulnerabilities associated with browsing the web, system penetration, information breach and identity threat; Securing the communication channel: encrypting data stream using SSL, confidentiality and integrity of data using third party transaction protocols e.g. SET, PCI DSS standard, the latest evolutions for HTTPS deployments; Securing untrusted data: server-side and client-side injection attacks, defending common injection attacks; Session management and access control: relationship between authentication, authorization and session management, prevent authorization bypasses and harden session management mechanisms; Server-side se-curity: CGI security, server configuration, access control, operating system security, malicious e-mails, web scripts, cookies, web bugs spyware, rogue AV etc.

CSE 4949: IT Audit: Concepts and Practice

Prerequisite course: CSE 3421

IT audit concepts and frameworks; General phases of IT audit; Internal IT audit control framework: the committee of sponsoring organizations (COSO); The impact of information technology audit process on internal controls: gen-eral controls, application controls, tests of controls; Referring case study; IT-Audit methodologies and frameworks: COBIT, ITIL, ISO 17799 etc; Practical IT-Audit methodologies development steps for enterprises completing the audit: reporting, types of auditors' opinions, audit documentation and resources; Referring case study.

CSE 3542: Industry Internship

Prerequisite Course: Completion of all credits except CSE 3542

A student must complete all their academic credits and requirements before they embarkment into internship program. Students are responsible for securing their own internships but should contact the Career Services Office / the Department Office for assistance and resources to identify and apply for opportunities of interest. Students will participate in an internship for a semester under a supervisor in industry and an examiner from the University. The students work at the industry will be evaluated through the daily journal and a standard rubric marked by the industry supervisor. This evaluation will be complemented by the evaluation of the students' work presented to the internship examiner in the department.

CSE 4435: Software Architecture

Introduction; Design vs Architecture; Enterprise Architecture; Architectural drivers; Software Architecture role; Skills and knowledge of software architect; Software architecture in the delivery process; Visualizing Software Architecture; Managing risks; Architectural recovery, architectural styles, domain specific software architectures coupled with programming/implementation effort, design and implement a real-world software system, the state-of-the-art in software architecture research and future trends.

CSE 4181: Mobile Application Development

Prerequisite Course: CSE 4165

Introduction to Mobile Programming; Mobile Programming Languages; Mobile Application Development Frame-work; MVC architecture; UI Architecture: activities and intents, activity life cycle, supporting multiple screen sizes; Working with data storage: storing and retrieving data, data persistence, SQLite database; User interface widgets: text controls, button controls, toggle buttons, images ; UI controllers: different types of dialog like alert, custom dialog, notification, showing Menu; Design patterns for handling synchronous/asynchronous tasks, processes and threads; View Animation; Network Communication: web services, handling HTTP requests and response, data parsing (JSON, XML); Mapping and location based services; Sensors: accelerometer, light sensor, microphone; Game development; Marketing; Distribution of Apps; Cross platform App development framework; Auto-Layout design.

CSE 4495: Software Testing and Quality Assurance

Prerequisite Course: CSE 3421

Testing strategies: SDLC vs STLC; Testing Levels; Testing methods; Testing types: Specification-based vs. code-based, black-box vs. white-box, functional vs. structural testing; unit, integration, system, acceptance, and

regression testing; Load, Performance, Stress, Unit Testing; Verification vs. validation; Test planning: scenario, case, traceability matrix; ISO Standards; Agile testing; Testing Estimation techniques; Introduction to software reliability, quality control and quality assurance; Formal verification methods; static and dynamic program verification.

CSE 4587: Cloud Computing

Prerequisite Course: CSE 3711, CSE 4509

Basic Concepts: cloud computing and applications, assessing the value proposition, issues and challenges, cloud architecture, service models, deployment models; Cloud Platforms: abstraction and virtualization, capacity planning, platform as a service, Amazon web services , Microsoft Azure, Google cloud platform; Cloud Infrastructure: managing the cloud, cloud security; Services and Applications: service-oriented architecture, moving applications to the cloud, cloud-based storage, media and streaming, cloud based mobile apps and web services.

CSE 4567: Advanced Database Management Systems

Prerequisite Course: CSE 3521

Database system architecture; Managing primary and secondary storage; Query processing; Metadata and catalog management; Language processing; Query optimization and plan generation; Concurrency; Failures and recovery; Extensibility; Client-server interactions; Object-oriented database systems, XML, database and the web, data management in distributed mobile computing environment, data broadcasting, text database, digital library design and implementation; Multimedia database: basic concepts, design and optimization of access strategies; Parallel database, spatial database, temporal database; Parallel and distributed database systems; NoSQL; New database architectures and query operators.

CSE 4413: Virtual Reality/Augmented Reality System Design

Prerequisite Course: CSE 3421

Introduction to Virtual Reality (VR) and Augmented Reality (AR), Fundamentals of immersive systems, VR/AR hardware components (HMDs, sensors, controllers), 3D graphics and rendering for immersive environments, Tracking and motion sensing technologies, Interaction techniques and user interface design for VR/AR, Development platforms and tools (e.g., Unity/Unreal), Spatial computing and environment mapping, Designing immersive user experiences, Prototyping VR/AR applications, Performance optimization and usability evaluation, Applications of VR/AR in gaming, education, healthcare, and industry, Ethical and social considerations of immersive technologies.

CSE 4611: Compiler Design

Prerequisite Course: CSE 2233

Compiler modules; Lexical analysis; Parsing theory; Symbol tables; Type systems; Scope; Semantic analysis; Intermediate representations; Runtime environments; Code generation; Code optimization.

CSE 4945: UI: Concepts and Design

Design principles: color, emphasis, usability, hierarchy, etc; Low fidelity wireframes: beginning to design using low fidelity wireframes and storyboards; Introduction to Sketch software, rapid prototyping using Sketch, high fidelity mobile, application, and website wireframes; Creating a style guide with Sketch; Design research and

personas: UX fundamentals; find, build, verify, patterns, personas, situations, buy-in, knowledge, scenarios; Using sketch to prototype using material design; Prototype employee time clock; Prototype tablet ordering interface; Prototype iOS todo app; Core principles of design: good, bad and ugly web search; Introduction to Illustrator, introduction to value: understanding Illustrator and designing in Illustrator, refactor and embellish, introduction to color with value, add hue to value; Introduction to PhotoShop, unity in design, PhotoShop and a UI tool, freeform of painting; Introduction to HTML and Visual Studio Code, learn markup language, tags and structure; Introduction to CSS, design guidelines, and styling; Styling with CSS. Complete content from CSS from scratch; Create new CSS on existing HTML; Basic site and app development in Bootstrap, develop a responsive site that will work on PCs tables and Phones; Basic site and App design in Bootstrap; Design graphics for the responsive site in the previous website.

CSE 4871: Data Analytics

Data analysis process; exploratory data analysis; selection of descriptive analytics techniques; identification of important attributes; development of derived attributes; predictive and prescriptive analytics; discovering patterns in datasets; data mining concepts and methodologies including classification, clustering, associations, and anomaly detection; selecting appropriate data mining algorithms.

CSE 4133: Business Intelligence

Prerequisite Courses: CSE 3411

Introduction to Business Intelligence concepts and evolution; data-driven decision making; data sources and data integration; data warehousing concepts and architectures; ETL (Extract, Transform, Load) processes; multidimensional data modeling; OLAP operations; data visualization and dashboards; descriptive and diagnostic analytics; basic predictive analytics; business reporting; performance management and key performance indicators (KPIs); ethical and governance issues in business intelligence; case studies on BI applications in business organizations.

CSE 4451: Human Computer Interaction

Foundations of human computer interaction: understanding and conceptualizing interaction; Understanding users: human perception, ergonomics, cognition, psychology; Task Analysis; User Interface Design, interface programming, graphical user interfaces, user survey, user journey and experience, mobile devices, multimodal interfaces and ubiquitous computing, user-centered system development and evaluation, user-centered software development and evaluation; Prototyping; Interaction design for new environments; Affective and social computing; Assistive and augmentative communication, assistive technology and rehabilitation; Human machine interface, brain computer interface; Experimental research ethics.

CSE 4485: Game Design and Development

Introduction to games: history, games and society; Game design: design concepts, teams and processes character modelling, animation, storyline, programming fundamentals, concepts of 3D virtual world; Game Engines: 3D mesh and object modelling, simulation and collision detection, etc; Debugging games; Game Architecture; Memory and I/O systems; Development of a customized game; Advanced Topics: data structures , AI, etc in Games; Networks and multiplayer mode; Application of Games: simulation, animation movies and others.

CSE 4519: Distributed Systems

Remote invocation and indirect communication; Time and coordination; Overlay networks and P2P; Distributed

storage and file systems; Name services; Global state and transactions; Replication and consistency; Consensus; Fault tolerance; Security and privacy; Emerging topics in distributed systems.

CSE 4523: Simulation and Modeling

Prerequisite Course: MATH 2205

Simulation methods, model building, random number generator, statistical analysis of results, validation and verification techniques; Digital simulation of continuous system; Simulation and analytical methods for analysis of computer systems and practical problems in business and practice; Introduction to simulation packages.

CSE 4621: Computer Graphics

Prerequisite Course: MATH 2183 MATH 2201

Basics of computer graphics and its applications; Raster graphics: 3D rasterization pipeline; Transformation: modelling, viewing and projection transformation in both 2D and 3D spaces; homogeneous coordinate system; Visible surface detection and hidden surface removal: e.g. z-buffer (or, depth buffer), depth-sorting, BSP-tree algorithms; Scan conversion and clipping algorithms: e.g. Cohen-Sutherland, Cyrus-Beck, Sutherland-Hodgman

Course Contents

algorithms; Fractals: e.g. Koch curve, Snowflakes, Dragon curve; Ray tracing: ray casting methods, direct illumination, global illumination, shadows, shading and textures.

CSE 4601: Mathematical Analysis for Computer Science

Prerequisite Course: MATH 2205, CSE 2213

Recurrent problems; Manipulation of sums; Integer functions; Number theory; Binomial coefficient; Special numbers; Generating functions; Combinatorial game theory; Introduction to probability theory, expectation; Random variables; Conditional Probability and Conditional Expectation; Stochastic process; Markov chains: discrete parameter, continuous parameter, birth-death process; Queuing models: birth-death model, Markovian model, open and closed queuing network; Application of queuing models.

CSE 4633: Basic Graph Theory

Prerequisite Course: CSE 2213, CSE 2217

Graphs and their applications; Basic graph terminologies; Basic operations on graphs; Graph representations; Degree sequence and graphic sequence; Paths, cycles and connectivity; Trees and counting of trees; Distance in graphs and trees; Spanning trees in graphs; Euler tours; Hamiltonian cycles; Ear decomposition; Graph labeling; Matching and Covering: Vertex and Edge Covering; Line graphs, Perfect graphs and Planar graphs; Graph coloring: Vertex coloring and Edge coloring; Special classes of graphs.

CSE 4655: Algorithm Engineering

Prerequisite Course: CSE 2217

Computational complexity; Exact Algorithms; Parameterized complexity; Practical computing and heuristics; Approximation algorithms; LP based approximation algorithms; Randomized algorithms; On-line algorithms; Experimental algorithmics; Contemporary and state-of-the-art algorithms.

CSE 4547: Multimedia Systems Design

Prerequisite Course: CSE 3711

Organization and structure of modern multimedia systems; text, audio and video encoding; Data compression: lossless and lossy techniques; Multimedia networking: Quality of Service management and multimedia protocols; Streaming multimedia: peer-to-peer, video-on-demand, live streaming; Multimedia storage: data placement and scheduling, caching, and data retrieval; Scheduling algorithms for multimedia within OS; Synchronization schemes: in-band and out-band, synchronization skews and specification; Design of real-world multimedia solution.

CSE 4613: Computational Geometry

Prerequisite Course: MATH 2201, CSE 2213

Searching and Geometric Data Structures: Balanced binary search trees, Priority-search trees, Range searching, Interval trees, Segment trees; Algorithms and complexity of fundamental geometric objects: Polygon triangulation and art gallery theorem, Polygon partitioning, Convex-hulls in 2-dimension and 3-dimension, Dynamic convex-hulls; Geometric intersection: Line segment intersection and the plane-sweep algorithm, Intersection of polygons; Proximity: Voronoi diagrams, Delunay triangulations, Closest and furthest pair; Visualization: Hidden surface removal and binary space partition (BSP) trees; Graph Drawings: Drawings of rooted trees (Layering, Radial drawings, HV-Drawings, Recursive winding), Drawings of planar graphs (Straight-line drawings, Orthogonal drawings, Visibility drawings).

CSE 4379: Real-time Embedded Systems

Prerequisite Course: CSE 4325

Embedded architectures: 16/32/64-bit embedded processors; Interaction with devices: buses, memory

architectures, memory management, device drivers; Concurrency: software and hardware interrupts, timers; Real-time principles: synchronization, scheduling, multi-tasking; Real-time task scheduling: schedulability analysis, rate and deadline monotonic scheduling, fixed and dynamic priority scheduling; Feedback control theory and application; Profiling and code optimization; Embedded software systems: exception handling, loading, mode-switching, programming embedded systems.

CSE 4337: Robotics

Prerequisite Course: CSE 4325, CSE 3811

Introduce the basic concepts of robotics, types of robots, robotics and AI; Automation & autonomy architectures; Robot hardware: sensors, actuators; Robotic mapping: localization, Monte Carlo localization, multi-object localization; Robotic navigation and locomotion: motion planning, dynamics and control; Human-robot interaction: Natural language learning; Multi-agents: tasks and teams.

CSE 4327: VLSI Design

VLSI technology: Top down design approach, technology trends and design styles. Review of MOS transistor theory: Threshold voltage, body effect, I-V equations and characteristics, latch-up problems, NMOS inverter, CMOS inverter, pass-transistor and transmission gates. CMOS circuit characteristics and performance estimation: Resistance, capacitance, rise and fall times, delay, gate transistor sizing and power consumption. CMOS circuit and logic design: Layout design rules and physical design of simple logic gates. CMOS subsystem design: Adders, multiplier and memory system, arithmetic logic unit. Programmable logic arrays. I/O systems. VLSI testing.

CSE 4853/EEE 4523: Internet of Things (IoT)

Prerequisite Courses: CSE 4325

Introduction to IoT: Definitions, layered architecture, applications, AI/ML on the edge, energy harvesting, nanorobotics, IoT device lifecycle; ESP-IDF: Platform installation, debugging; Embedded hardware essentials: GPIO, digital/analog I/O, PWM, timers, ADC/DAC, power management; Protocols for Sensors: UART, I²C, SPI, Modbus; Actuators: Relays, Motors; Low power applications: Waking up from sleep, ULP coprocessor in deep sleep; Outputs with display: LCDs, OLEDs, TFTs; RTOS and structured firmware: Tasks, queues, watchdog timers; Networking fundamentals: LAN vs WAN, IP addresses, MAC addresses, DNS. DHCP; Light weight IP: Basics of lwIP, mDNS, SNTP; IoT device security basics: Common IoT vulnerabilities, default passwords, insecure update, open debug ports; End-to-end IoT security: TLS tunnel, device identity (certificates PKI), secure OTA updates with rollback, RPC; Wireless IoT connectivity: Station vs Access Point modes, SSID scanning; Bluetooth Low Energy Connectivity: Basics, developing a BLE beacon, Developing a GATT server, Setting up BLE mesh network; Internet protocols in IoT: TCP, UDP, HTTP basics; Publish/Subscribe messaging: Messaging model for IoT; broker, topics, Quality of Service; Lightweight protocols: REST-like communication for constrained devices (CoAP); Low-power wide-area networks (LPWAN): Long-range IoT networks; Cloud IoT platforms: Adafruit IoT, ThingsBoard IoT, AWS IoT, Azure IoT, etc; Guided project implementation: Choosing project domain, Implementation, Report submission, Presentation.

CSE 4819: AI for Autonomous Systems

Prerequisite Courses: CSE 4889

Introduction to autonomous systems and autonomy levels; foundational concepts of artificial intelligence and

intelligent agents; perception and environment sensing (e.g., camera, lidar, radar); sensor fusion and state estimation; mapping and localization techniques; path planning and decision-making algorithms; motion control and navigation; machine learning methods for autonomy (supervised, unsupervised, reinforcement learning); deep learning for perception and control; simulation environments for autonomous agents; safety, robustness, and ethical considerations; case studies on autonomous vehicles, drones, mobile robots, and industrial autonomous systems.

CSE 4813/DS 4211: Deep Learning

Prerequisite Courses: CSE 4889

Foundations of Neural Networks and Deep Learning: components of a learning algorithm, activation functions, loss functions, back propagation, multi-layer perceptron, regularization, dropouts, weight decay, batch normalization, optimization algorithms; Convolutional Neural Networks: convolution and pooling, variants of convolutional layers, dilated convolution, transfer learning; Recurrent Neural Networks: computing gradients in RNN, deep RNN; sequence-to-sequence architectures, word embedding, recursive networks, backpropagation through time, vanishing and exploding gradients, long short term memory, self attention, transformer; Deep Unsupervised Learning: autoencoders, variational autoencoders, generative adversarial networks.

CSE 4815: Large Language Model and Generative AI

Prerequisite Courses: CSE 4811/DS 4213

Introduction to Large Language Models and Generative AI; History and evolution of language models; Transformer architecture: self-attention mechanism, multi-head attention, positional encoding, encoder-decoder vs decoder-only architectures; Data wrangling and tokenization methods; Training LLMs: distributed training, scaling strategies, compute optimization; Pre-training and foundation models: MLM, NSP, pretraining objectives; Fine-tuning techniques: supervised fine-tuning (SFT), transfer learning; Adaptation strategies: LoRA, adapters, prefix tuning, instruction tuning, RLHF; Efficient LLMs: knowledge distillation, quantization, model pruning; Evaluation metrics and benchmarks: GLUE, SuperGLUE, MMLU; Applications: text generation, summarization, question answering, code generation; Ethical considerations: bias types, misinformation, privacy implications, responsible AI frameworks; Security: prompt injection, jailbreak attacks, data poisoning, adversarial attacks; Multimodal LLMs: vision-language models, text-to-image generation (DALL-E, Stable Diffusion), visual question answering; Future directions: retrieval-augmented generation (RAG), Future Applications of LLMs.

CSE 4329: Digital System Design

Prerequisite Course: 3313

Design using MSI and LSI components; Programmable logic devices; Basic components of a computer system; Design of processing unit: ALU, Comparator, Accumulator, Shifter, Multiplier; Hardware multiplication: Booth and Modified Booth algorithm; Design of control unit: hardwired and microprogrammed; Simple-As-Possible (SAP) computer: SAP-1, selected concepts from SAP-2 and SAP-3 (jump, call, return, stack, push and pop); Designing microprocessor based system; Hardware Interfacing with Intel 8086 microprocessor: programmable peripheral in-terface, programmable interrupt controller, programmable timer, keyboard and display interface.

CSE 4397: Interfacing

Prerequisite Course: CSE 4325

Definition of interface, types of interfaces; Interface levels; Typical interface mechanisms; Example interfaces; In-put/output ports: I/O port structure, status and control data registers, bidirectional pin operation, bus connection; Three-state output, Z state; Technological considerations; Connections to external loads; Input device connections; Signal multiplexing; Analog Interfaces; Timing and frequency aspects of analogue signals; Nyquist-Shannon sampling theorem; Analog-digital converters; Digital-analog converters; Example application; Serial communication interfaces; Types and characteristics of communication interfaces; Synchronous serial interface (SSI). Real examples (RS232, SPI); Common computer Interfaces; Universal Serial Bus (USB), USB3, Thunderbolt PCI express (PCIe), Storage interfaces – SATA, NVMe, eMMC; Display interfaces – VGA, DVI, Display Port; Microcontroller interfacing – Arduino, Raspberry pi GPIO; Buses and DMA; Design and operation of interface between computer and the outside world; Human computer interaction, brain Computer interfaces.

CSE 4891: Data Mining

Prerequisite Course: CSE 4889

Introduction to data mining: data mining task and applications, data preprocessing, feature selection, association analysis, frequent item-set mining; Single model classifier: k-nearest neighbor, naïve Bayes classifier, decision tree induction, naïve Bayesian tree, rule-based classifiers; Model evaluation and selection; Ensemble learning: random Forests, bagging, boosting, isolated forests; Clustering: k-means clustering, similarity-based clustering, nearest-neighbor clustering, density-based clustering, ensemble clustering, evaluation of clustering methods, clustering high-dimensional data; Data balancing methods; Active learning; Transfer learning; Outlier detection; Concept drift.

CSE 4817: Big Data Analytics

Prerequisite Course: CSE 4889

Introduction to Big Data: characteristics of Big Data and dimensions of scalability; Data Science: getting value out of Big Data, foundations for Big Data systems and programming, getting started with Hadoop; Big Data Modelling and Management Systems: Big Data modelling, Big Data management, designing a Big Data management system; Big Data Integration and Processing: retrieving Big Data, Big Data integration, processing Big Data, Big Data analytics using Spark; Machine Learning with Big Data: introduction to machine learning with Big Data, data exploration, classification, evaluation of machine learning models, regression, cluster analysis, and association analysis; Graph Analytics for Big Data: introduction to graphs, graph Analytics, graph analytics techniques, computing platforms for graph analytics.

CSE 4823: Data Driven Decision Making

Prerequisite Courses: CSE 4133

Exploring data, Probability and decision making under uncertainty, statistical inference, regression analysis and time series forecasting, optimization and simulation modeling, advanced data analysis.

CSE 4821/MKT 4313 : Digital Marketing

Digital Marketing Essentials: The Digital Marketing Environment; The Digital Consumer; Content Marketing; Digital Marketing Tools and Channels: Social Media Marketing; Augmented, Virtual and Mixed Reality, Search Engine Optimization (SEO); Search Engine Marketing (SEM); Digital Marketing Strategy and Planning: Digital Marketing Strategy and Objectives; Digital Marketing Management: Integrating, Improving and Transforming Digital Marketing; Further issues in legal and social Sensitivity.

CSE 4941: Enterprise Systems: Concepts and Practice

Materials management (MM); Supply chain management (SCM); Customer relationship management (CRM); Financials, mobile and cloud enterprise systems; Internet-of-Things (IoT) and enterprise BIG data; The course will incorporate a hands-on component using SAP, Oracle ES software. The course will also

incorporate modelling techniques and tools, assess an organisation's readiness for ES implementation.

CSE 4763: Electronic Business

Prerequisite Course: CSE 3711

The E-Business Framework: difference between electronic business and electronic commerce, electronic markets, disintermediation, horizontal and vertical market places; E-Products and E-Services; Classification of business webs: agora, aggregation, value chain, alliance, supply chain net; business model for e-products and e-services, branding and pricing; E-Procurement: difference between purchase and procurement, market solutions: sell-side, buy-side, and market place; Integration of product catalogue, procurement service providing; Online Marketing: comparison of online media, usage of Internet and websites, stages of a customer development model: surfer, consumer, prosumer, buyer, and key customer; E-Contracting: generic services, information, negotiation, archiving, enforcement, reconciliation, structure of a contract, digital signature, legal affairs; Online Distribution: components of a distribution system, characterisation of online distribution, hybrid distribution networks, model for electronic software distribution; E-Payment: electronic means of payment, micro and macro payment, classification of payment systems, credit cards, customer accounts, digital money; secure transactions; Electronic customer relationship management: objectives of CRM, customer acquisition and liaison, customer buying cycle, architecture of CRM systems, customer satisfaction survey; E-Business environment: information society, building process for communities, multi-option society, ethics in electronic business.

CSE 4811/DS 4213: Natural Language Processing

Prerequisite Courses: CSE 4889

Word representations: Representing documents for neural networks, Introduction to embeddings. Continuous bag-of-words representations; Pre-training word embedding models: introduction, SGNS algorithm, Evaluation and interpretation of word embeddings, Subword representations; Language modelling: N-gram language models, Recurrent neural networks, The LSTM architecture, Recurrent neural network language models, Contextualized word embeddings; Introduction to generation tasks, Evaluation of generation systems; Introduction to machine translation, Neural machine translation; Attention, The Transformer architecture; Pre-trained transformer models: GPT, BERT; Summarization; Dialogue Systems; Domain Adaptation; Structured prediction.

CSE 4893: Introduction to Bioinformatics

Prerequisite Course: CSE 4889

Introduction; Molecular biology basics: DNA, RNA, genes, and proteins; Graph algorithms: DNA sequencing, DNA fragment assembly, Spectrum graphs; Sequence similarity; Suffix Tree and variants with applications; Genome Alignment: maximum unique match, LCS, mutation sensitive alignments; Database search: Smith-Waterman algorithm, FASTA, BLAST and its variations; Locality sensitive hashing; Multiple sequence alignment; Phylogeny reconstruction; Phylogeny comparison: similarity and dissimilarity measurements, consensus tree problem; Genome rearrangement: types of genome rearrangements, sorting by reversal and other operations; Motif finding; RNA secondary structure prediction; Peptide sequencing; Population genetics; Recent Trends in Bioinformatics.

CSE 4883: Digital Image Processing

Prerequisite Course: CSE 4889

Digital Image Fundamentals: visual perception, sensing, acquisition, sampling, quantization; Intensity Transformation and Spatial Filtering: different transformations, histogram, correlation and convolution,

smoothing and sharpening filters; Filtering in Frequency Domain: discrete-fourier-transformation (DFT) of image, smoothing and sharpening in frequency domain, selective filtering; Image Restoration and Reconstruction: noise models, spatial filtering for noise, frequency filtering for noise, reconstruction from projections;

Color Image Processing: color models, color transformation and segmentation;

Morphological Image Processing: erosion, dilation, opening, closing, morphological algorithms; Image Compression: redundancy, fidelity criteria, some basic compression techniques; Image Segmentation: point, line and edge detection, thresholding, region based segmentation; Object Recognition: matching, statistical classifier, neural networks.

CSE 4825: Speech Processing and Recognition

Prerequisite Courses: CSE 4889

Introduction to speech and audio signals; human speech production and perception; basic digital signal processing concepts for speech; speech signal acquisition and preprocessing; time-domain and frequency-domain analysis; short-time analysis of speech signals; feature extraction techniques including MFCC, LPC, PLP, and spectrograms; speech modeling techniques; statistical models for speech recognition; Hidden Markov Models (HMMs) for speech; introduction to deep learning approaches for speech recognition; acoustic modeling and language modeling; speech recognition systems and architectures; speaker recognition and verification; speech synthesis overview; noise reduction and robust speech processing; evaluation metrics for speech recognition systems; applications and case studies in automatic speech recognition (ASR), voice assistants, and speech-enabled systems.

Final Year Design Project

Final Year Design Project (FYDP) is a senior design project work that takes place during the final year of 4 years engineering curriculum of B.Sc. in Computer Science and Engineering.

CSE 4000A: Final Year Design Project - I

This course introduce different soft skill-sets that are necessary for the successful completion of FYDP. The skill-sets include, but not limited to, mastering effective communications, individual and team development, ethical leadership, project management, the steps in the design process, environment and sustainability, etc. These skill sets would be developed by a series of seminars and workshops. The outcomes relevant to POs would be measured based on the student performance in different tests designed to assess those specific skills. The standard rubrics will be used to assess the performance. At the end of the trimester the students will submit an interim report of their FYDP and give a presentation.

CSE 4000B: Final Year Design Project - II

In this course, the students will implement the proposal that is accepted in the course CSE 4000A.

CSE 4000C: Final Year Design Project - III

In this course, the students will implement the proposal that is accepted in the course CSE 4000A.

Implementation Schedule:

Effective from the students of Summer 2026 trimester.

For batches before Summer 2026:

Students can select the majors from the new curriculum (Five Majors) if they complete their respective curriculum and complete the requirements of the respective major.